PRODUCT SPECIFICATION DOCUMENT · CONFIDENTIAL

Prism

Bloomberg meets Product Hunt — the full arc of tech product history. Signal scoring, attribute taxonomy, knowledge graph, market analytics.

14,114	25	11	$47
PRODUCTS IN DB	ROUTES PLANNED	ROUTES BUILT	MONTHLY INFRA

April 2026 · Codename: prism · Repo: alimabsoute/v0-product-sentinel

Status Overview

All 25 planned routes — current build status at a glance.

PROGRESS BY CATEGORY

Core intelligence pages	35%
Analytics & markets	50%
Taxonomy browsing	15%
Data pipeline & signals	55%
Auth & user features	30%
Monetization	0%

ALL ROUTES

`/`

Home Feed PARTIAL

Layout built. Missing: breakout alerts wired, funding feed, graveyard strip, category spotlight chart.

`/products`

Product Browse PARTIAL

Basic category filter works. Missing: 10-attribute faceted sidebar, era filter, sub-category drill.

`/dossier/[slug]`

Product Dossier NOT BUILT

Still at /products/[slug]. Needs rename + full intelligence layout: signal charts, funding timeline, relationships graph, alts.

`/markets`

Market Analytics PARTIAL

Basic charts built. Missing: category heatmap, attribute market share over time, acquisition tracker, breakout history.

`/functions/[slug]`

Function Pages NOT BUILT

Leaf taxonomy pages (e.g. /functions/markdown-note-editor). $-chart, lifecycle, top 10 ranked.

`/tools-to/[task]`

Task SEO Pages NOT BUILT

TAAFT-style discovery. Powers long-tail SEO. Depends on task_search_tags being populated.

`/compare`

Compare View PARTIAL

Basic side-by-side built. Missing: radar overlay, attribute diff panel, AI verdict (Claude), funding comparison.

`/explore`

Knowledge Graph BUILT

ForceGraph2D with hover, click panel, search. Edges sparse until product_relationships populated.

`/evolution`

Era Timeline PARTIAL

Category growth + death timeline built. Missing: era swim-lanes, interactive timeline dots, hall of fame.

`/graveyard`

Graveyard PARTIAL

Timeline layout built. Status filter just fixed. Missing: death reason donut, pre-death signals teaser (paid CTA).

`/categories/[slug]`

Category Pages BUILT

Filter by status, search within category. Missing: lifecycle chart, sub-category drill-down.

`/insights + /[slug]`

Insights Hub PARTIAL

Article reader exists. Missing: Obsidian-style graph sidebar, signal chart rail, related product cards.

`/companies/[slug]`

Company Profile NOT BUILT

All products, funding history, valuation chart, related companies.

`/trending`

Breakout Feed NOT BUILT

Real-time is_breakout = TRUE feed with sparklines + velocity chart. Data

`/attributes/[grp]/[val]`

Attribute Slices NOT BUILT

E.g. /attributes/deployment/self-hosted — all self-hosted products.

Needs companies table populated.

exists, page missing.

Depends on product_tags populated.

`/funding`

Funding Feed NOT BUILT

Funding round chronological feed. funding_rounds table empty — needs ingestion wired.

`/acquisitions`

M&A Tracker NOT BUILT

acquisitions table not created yet. Phase 3+ feature.

`/profile`

User Profile PARTIAL

Auth wired (Supabase). Watchlist UI needs user_saves populated. Alert system not built.

`/login + /signup`

Auth Pages BUILT

Supabase auth fully wired. Suspense boundaries fixed.

`/submit`

Product Submit NOT BUILT

User-submitted product form. Phase 4+ feature.

`/pricing`

Pricing Page NOT BUILT

Free / Pro / API tiers. Monetization phase.

`/api/docs`

API Docs BUILT

Static docs page exists.

`/search`

Unified Search NOT BUILT

⌘K palette exists but no dedicated /search results page. pg_trgm + pgvector hybrid ready.

`/sub-categories/[slug]`

Sub-category Pages NOT BUILT

Mid-level taxonomy rollup. 100+ sub-categories in DB, no page yet.

`/watchlist`

Watchlist + Alerts PLANNED

user_saves + user_alerts tables exist (RLS). UI not connected. Phase 4+.

/ — Home Feed

First page users see. Product Hunt-style present-tense launch feed with Bloomberg signal overlays.

WIREFRAME

```
┌──────────────────────────────────────────────────────────┐
│ HERO: "47 new products today · 8 breakouts"              │
│        [Submit your product →]                           │
├──────────────────────────────┬───────────────────────────┤
│  🚀 LAUNCHED TODAY           │  📊 MARKET PULSE          │
│  ┌──────┬──────┬──────┬──────┐│ AI Tools    ▲ +12%      │
│  │ card │ card │ card │ card ││ Dev Tools   ▲ +8%       │
│  ├──────┼──────┼──────┼──────┤│ Prod        —  0%       │
│  │ card │ card │ card │ card ││ Design      ▼ -2%       │
│  └──────┴──────┴──────┴──────┘│ [markets →]             │
│  [view all launches →]       ├───────────────────────────┤
│                              │ 📰 FROM THE NEWS          │
│  ⚡ BREAKOUT ALERTS           │ · TechCrunch 2h           │
│  [logo] Product X  +340% ▲   │   "Notion raises ... "    │
│  [logo] Product Y  +220% ▲   │ · The Verge  4h           │
│  [logo] Product Z  +180% ▲   │ · HN         6h           │
│  [view all breakouts →]      ├───────────────────────────┤
│                              │ 💰 FRESH FUNDING          │
│  📈 TRENDING THIS WEEK       │ $50M Series B → X         │
│  1. Supabase    +67 pts ▲    │ $12M Seed    → Y          │
│  2. Resend      +54 pts ▲    │ Acquired Z   → BigCo      │
│  3. Turso       +48 pts ▲    ├───────────────────────────┤
│                              │ 🪦 GRAVEYARD              │
│  🏛 CATEGORY SPOTLIGHT       │ Product X   3d ago        │
│  "Note-taking peaked 2019"   │ Product Y acquired        │
│  [area chart 1990→2026]      │                           │
└──────────────────────────────┴───────────────────────────┘
```

FEATURE STATUS

WIDGET	STATUS
Hero strip (stats)	BUILT
Launched Today grid	BUILT
Market Pulse sparklines	PARTIAL
Breakout Alerts feed	MISSING
From the News feed	PARTIAL
Trending This Week	BUILT
Fresh Funding strip	MISSING
Category Spotlight chart	MISSING
Graveyard this week	MISSING

WHAT NEEDS TO HAPPEN

1. **Wire Breakout Alerts to live data** — P0 · S · 2h
   Query `product_signal_scores WHERE is_breakout = TRUE AND score_date = today`. Run `pnpm signals` first to score all 14K products. Currently 6,275 scored.

2. **Add Category Spotlight chart** — P1 · M · 4h
   Rotating lifecycle area chart per category. Query `category_monthly_launches` view (already exists in migration 0006). Pick a random active category on each render.

3. **Add Fresh Funding strip** — P2 · L · 1d
   Wire to `funding_rounds ORDER BY year,month DESC LIMIT 6`. Table exists but is empty — needs funding ingestion from press_mentions event_type='funding'.

4 **Add Graveyard this week strip** P2 S · 1h

Simple query on recently discontinued products. Already fixed the status filter bug (discontinued/inactive). Just needs a small widget component on home page.

/dossier/[slug] — Full Product Intelligence File

THE monetizable page. Everything known about one product. Long-scroll with sticky TOC. Free tier: 10/mo. Pro: unlimited.

WIREFRAME

```
┌──────────────────────────────────────────────┐
│ ┌────┐  PRODUCT NAME          [🚀 Save][📤 Share] │
│ │LOGO│  Tagline in one sentence                   │
│ └────┘  [AI Tools][Note-Taking][Markdown-Editor] │
│         🔗 website   🐙 github   🐦 twitter      │
├──────────────────────────────────────────────┤
│ Age:6yr │ $280M │ Signal:87▲ │ BREAKOUT │ 5/5   │
├──────────────┬───────────────────────────────┤
│ STICKY TOC   │ 📝 DESCRIPTION + SCREENSHOTS    │
│ • Overview   │ [carousel img][carousel img]    │
│ • Attributes │                                 │
│ • Scores     │ 🏷️ ATTRIBUTE PANEL              │
│ • Signals    │ Capability: [collab][real-time] │
│ • Funding    │ Audience:   [devs][solopreneurs] │
│ • Revenue    │ Pricing:    [freemium]          │
│ • Press      │ Deployment: [cloud][desktop]    │
│ • Relations  │ UX:         [cmd-palette]       │
│ • Alts       │                                 │
│ • Changelog  │ 📊 DIMENSIONAL RADAR            │
│ • Tasks      │ [radar chart - 8 dimensions]    │
│ • Insights   │ Ease:4 Depth:5 Value:3 Perf:5   │
│ • Metadata   │                                 │
│              │ 🔥 SIGNAL CHARTS (2×2)          │
│              │ ┌──────────┬──────────┐         │
│              │ │Mentions  │Sentiment │         │
│              │ ├──────────┼──────────┤         │
│              │ │Velocity  │Platform  │         │
│              │ └──────────┴──────────┘         │
│              │                                 │
│              │ 💰 FUNDING TIMELINE             │
│              │ 2019●Seed$2M → 2022●B$50M       │
│              │                                 │
│              │ 📰 PRESS MENTIONS (feed)        │
│              │    RELATIONSHIPS (mini graph)   │
│              │ 🔄 ALTERNATIVES (card grid)     │
│              │ 📜 CHANGELOG / HISTORY          │
│              │ 🎯 RELATED TASKS                │
│              │ ℹ️ DATA METADATA                │
└──────────────┴───────────────────────────────┘
```

FEATURE STATUS

SECTION	STATUS
Route exists at /products/[slug]	BUILT
Rename to /dossier/[slug]	MISSING
Hero + quick stats strip	PARTIAL
Description + screenshots	BUILT
Attribute panel (Layer 2 tags)	PARTIAL
Dimensional radar chart	MISSING
Signal charts 2×2 grid	MISSING
Funding timeline chart	MISSING
Press mentions feed	PARTIAL
Relationships mini graph	MISSING
Alternatives card grid	PARTIAL
Sticky TOC navigation	MISSING
Changelog / history	MISSING
Related tasks → /tools-to/	MISSING
Data freshness metadata	MISSING

WHAT NEEDS TO HAPPEN

1. **Populate product_tags for all 14K products** — P0 · XL · 1d
   BLOCKER for attribute panel, alternatives, /functions pages, /tools-to pages, and /products faceted sidebar. Run attribute extraction pass via Claude Haiku (~$14 one-time cost for 14K products).

2. **Add dimensional radar chart** — P1 · M · 3h
   Populate `functionality_scores` JSONB during extraction pass above. Render with recharts `RadarChart` (already installed). 8 dimensions: ease_of_use, feature_depth, value, support, performance, docs, mobile, integrations.

3 **Add signal charts 2×2 grid** P1 M · 4h

Mention volume (stacked bar by platform), sentiment trend (rolling line), velocity WoW (line), platform mix (donut). Data in `product_signal_scores`. Needs `social_mentions` table populated for platform breakdown.

4 **Add sticky TOC + funding timeline** P2 M · 4h

Sticky rail requires CSS `position: sticky` + IntersectionObserver for active section. Funding timeline chart needs `funding_rounds` populated via news extraction pipeline.

5 **Rename route /products/[slug] → /dossier/[slug]** P2 S · 2h

Update 6 files: product-card.tsx (4 variants), search-command.tsx, news-feed.tsx. Add redirect from old URL. Update "Open dossier" CTA copy.

/markets — Bloomberg-Style Analytics

$ aggregations across categories, attributes, functions, time. This is THE moat — nobody else has it.

WIREFRAME

```
┌──────────────────────────────────────────────────────────┐
│ MARKETS   [Category▾][Attribute▾][Time▾][CSV]            │
├──────────────────────────────────────────────────────────┤
│  📊 CATEGORY HEATMAP (full width)                        │
│  ┌──────────────────────────────────────────────────┐   │
│  │      2019  2020  2021  2022  2023  2024          │   │
│  │ AI   ▒▒▒   ▒▒▒   ███   ███   ███   ███           │   │
│  │ Dev  ▒▒▒   ███   ███   ▒▒▒   ▒▒▒   ███           │   │
│  │ Prod ▒▒▒   ███   ▒▒▒   ▒▒▒   ▒▒▒   ▒▒▒           │   │
│  │ (color = VC $ invested · tooltip shows $)        │   │
│  └──────────────────────────────────────────────────┘   │
├──────────────────────────┬───────────────────────────────┤
│ 💰 FUNDING LEADERBOARD   │ 📊 LIFECYCLE CURVES           │
│ 1. AI Tools    $4.2B ▲▲▲ │ ┌──────────────────┐          │
│ 2. Dev Tools   $1.8B ▲▲  │ │ Notes ──/\──     │          │
│ 3. Fintech     $1.5B ▲   │ │ CRM   ──__/──    │          │
│ 4. Security    $900M     │ │ WordP  ───\__    │          │
│ 5. Design      $400M ▼   │ │ 1990 2000 2010   │          │
│                          │ └──────────────────┘          │
├──────────────────────────┼───────────────────────────────┤
│ 📈 SURVIVAL CURVES       │ 🎯 ATTR. MARKET SHARE         │
│ "What % of 2018 cohort   │ ▇▇ open-source                │
│  still alive in 2026?"   │ ▒▒ proprietary                │
│ 2015 ──\                 │ stacked area 2010→2026        │
│ 2018 ───\────            │ [toggle attribute ▾]          │
│ 2022 ────\__             │  → api-first                  │
│                          │  → self-hosted                │
│                          │  → ai-assist                  │
├──────────────────────────┴───────────────────────────────┤
│  🛒 ACQUISITION TRACKER (table)                          │
│  Figma · Adobe · $20B · 2024 · integrated                │
│  Product X · BigCo · $500M · 2024 · shutdown             │
├──────────────────────────────────────────────────────────┤
│  ⚡ BREAKOUT HISTORY (time-series bar chart)             │
│  |  | ||  || |  |||| |  ||                              │
│  ──────────────────────── Jan Feb Mar Apr                │
└──────────────────────────────────────────────────────────┘
```

FEATURE STATUS

WIDGET	STATUS
Stat cards (totals)	BUILT
Category lifecycle curves	BUILT
Signal distribution chart	BUILT
Velocity leaders table	BUILT
Survival rates chart	BUILT
Category heatmap	MISSING
Funding leaderboard	MISSING
🎯 Attribute market share The "components over time" feature you asked about	MISSING
Acquisition tracker table	MISSING
Breakout history bar chart	MISSING
Filter bar (cat/attr/time)	MISSING
CSV export	MISSING

WHAT NEEDS TO HAPPEN — ATTRIBUTE MARKET SHARE (THE FEATURE YOU ASKED ABOUT)

1. **Populate product_tags (the root blocker)** — P0 · XL · 1d
   The attribute market share chart requires product_tags to be populated. Currently mostly empty. Run attribute extraction pass on all 14K products via Claude Haiku.

2. **Build the query + chart** — P1 · M · 4h
   SQL: `SELECT launched_year, tag_group, slug, COUNT(*) FROM products JOIN product_tags JOIN tags GROUP BY launched_year, tag_group, slug`. Render as recharts `AreaChart` with stacked series per attribute value. Dropdown to switch between tag_groups.

3. **Populate funding_rounds for leaderboard + heatmap** — P1 · L · 1d

Requires extracting funding data from press_mentions (event_type='funding'). Already have the extraction prompt designed. ~$3/mo ongoing with Haiku.

4 **Build acquisition tracker** P2 L · 1d

New `acquisitions` table needed. Seed from press_mentions where event_type='acquisition'. Then wire paginated table to the page.

/products — Faceted Browse

The search entry point. 10-attribute group sidebar. The full 4-layer taxonomy exposed as filters.

WIREFRAME

```
┌──────────────────────────────────────────────────┐
│ PRODUCTS  [🔍 search]  Sort:Buzz▾  ▦Grid  ≡List  │
├──────────────────────┬───────────────────────────┤
│                      │                           │
│  🎛 FILTERS          │ Active: [AI Tools×][frm×] │
│                      │                           │
│  LAYER 1             │ ┌────┬────┬────┬────┐     │
│  Category ▾          │ │card│card│card│card│     │
│   ☑ AI Tools         │ ├────┼────┼────┼────┤     │
│   ☐ Dev Tools        │ │card│card│card│card│     │
│   ☐ Productivity     │ ├────┼────┼────┼────┤     │
│  Sub-category ▾      │ │card│card│card│card│     │
│   ☐ Code Editors     │ └────┴────┴────┴────┘     │
│   ☐ Note-Taking      │                           │
│   ☐ Vector DBs       │ ←  1  2  3 … 119  →       │
│                      │                           │
│  LAYER 2 ATTRS       │                           │
│  Capability ▾        │                           │
│   ☐ collaborative    │                           │
│   ☐ real-time        │                           │
│   ☐ ai-assist        │                           │
│   ☐ api-first        │                           │
│  Audience ▾          │                           │
│   ☐ developers       │                           │
│   ☐ enterprise       │                           │
│  Deployment ▾        │                           │
│   ☐ cloud            │                           │
│   ☐ self-hosted      │                           │
│   ☐ browser-ext      │                           │
│  Pricing ▾           │                           │
│  Tech Stack ▾        │                           │
│  UX Pattern ▾        │                           │
│  Compliance ▾        │                           │
│  Integrations ▾      │                           │
│  Business Model ▾    │                           │
│                      │                           │
│  ERA                 │                           │
│  ☐ mainframe         │                           │
│  ☐ pc era            │                           │
│  ☐ web1              │                           │
│  ☐ mobile            │                           │
│  ☑ ai era            │                           │
└──────────────────────┴───────────────────────────┘
```

FEATURE STATUS

FEATURE	STATUS
Product grid + pagination	BUILT
Category filter pills	BUILT
Sort by buzz/trending/az	BUILT
Text search (pg_trgm)	BUILT
10-attribute sidebar filters	MISSING
Sub-category filter	MISSING
Era filter	MISSING
Active filter chips (remove)	MISSING
List view toggle	MISSING

WHAT NEEDS TO HAPPEN

1. **Populate product_tags (P0 blocker)** — P0 · XL
   Nothing in the sidebar works without tags. Same extraction pass as dossier.

2. **Build filter sidebar component** — P1 · L · 1d
   shadcn Accordion + Checkbox per tag_group. URL query params drive filter state (shareable URLs).

3. **Extend searchProducts() for attribute filters** — P1 · M · 3h
   Add `tags?` param to `SearchParams` type, join through `product_tags` in the query.

Missing Pages — Taxonomy, SEO & Discovery

These pages are unbuilt but were designed from day 1. Together they form the discovery + SEO engine.

/FUNCTIONS/[SLUG]

E.g. /functions/markdown-note-editor — all products that do one specific thing.

```
┌──────────────────────────┐
│ Markdown Note Editor     │
│ 214 products · Prod > NT │
├──────────────────────────┤
│ 📈 Lifecycle chart       │
│    peak: 2019 (312 prods)│
├──────────────────────────┤
│ 💰 $1.2B total VC        │
├──────────────────────────┤
│ 🏆 Top 10 ranked         │
│ 1. Notion      87▲       │
│ 2. Obsidian    82        │
│ 3. Craft       79▲       │
├──────────────────────────┤
│ [4×6 product grid]       │
└──────────────────────────┘
```

NOT BUILT P1

Needs: functions table seeded (~400 values), primary_function_id FK on products, product_tags populated.

/TOOLS-TO/[TASK]

E.g. /tools-to/write-cold-emails — TAAFT-style task discovery + SEO goldmine.

```
┌──────────────────────────┐
│ Tools to:                │
│ "Write cold emails"      │
│ 18 products found        │
├──────────────────────────┤
│ 1. [logo] Apollo.io      │
│    Signal: 91▲           │
│    [AI][CRM][freemium]   │
├──────────────────────────┤
│ 2. [logo] Instantly.ai   │
│ 3. [logo] Clay           │
│ 4. [logo] Lemlist        │
│ ...                      │
└──────────────────────────┘
```

NOT BUILT P1

Needs: task_search_tags populated on products. Already `TEXT[]` column — just needs extraction pass.

/ATTRIBUTES/[GROUP]/[VALUE]

E.g. /attributes/deployment/self-hosted — every self-hosted product ranked.

```
┌──────────────────────────┐
│ Self-Hosted Products     │
│ deployment · 341 products│
├──────────────────────────┤
│ Growth trend ▲ +180%     │
│ since 2020               │
│ [area chart 2018→2026]   │
├──────────────────────────┤
│ [product grid]           │
└──────────────────────────┘
```

NOT BUILT P2

Needs: product_tags populated. Dynamic route straightforward once data is in.

/TRENDING — BREAKOUT FEED

```
┌──────────────────────────────────┐
│ TRENDING NOW · 8 breakouts today │
│ [Global velocity chart - 30d]    │
├──────────────────────┬───────────┤
│ [logo] Product X     │ +340% ▲▲▲ │
│ velocity > 2σ avg    │ [sparkline]│
│ Reddit + HN surge    │ Signal: 94 │
├──────────────────────┼───────────┤
│ [logo] Product Y     │ +220% ▲▲  │
│ funding + press      │ Signal: 88 │
└──────────────────────┴───────────┘
```

NOT BUILT P0

Easy win. Data already exists in `product_signal_scores` `WHERE is_breakout = TRUE`. Just needs a page. ~2h build.

/COMPANIES/[SLUG] — COMPANY PROFILE

```
┌──────────────────────────────────┐
│ [logo] Notion Labs               │
│ $280M raised · 500 employees     │
├──────────────────────────────────┤
│ Products:                        │
│ [Notion][Notion AI][Notion Calendar] │
├──────────────────────────────────┤
│ 💰 Funding timeline              │
│ 2019●$10M → 2021●$275M           │
├──────────────────────────────────┤
│ Related companies (co-invested)  │
└──────────────────────────────────┘
```

NOT BUILT P2

Needs: `companies` table populated (partially empty now), funding_rounds with company_id. Phase 2+.

Data Pipeline — What's Missing

The UI is only as good as the data behind it. These are the infrastructure gaps blocking features.

CURRENT DB STATE

Products	Signal Scores	Press Mentions	Product Tags
14,114	6,275	73	~0
14,103 active	44% coverage	RSS accumulating	Critical gap

Funding Rounds	Relationships	Alternatives	Social Mentions
0	0	0	0
Not ingested	Script ready	Script ready	Table not created

PIPELINE PRIORITY LADDER

1. **Attribute extraction pass — populate product_tags** — P0 · XL · 1d
   CRITICAL BLOCKER. Unlocks: /products faceted sidebar, /functions pages, /tools-to pages, /attributes pages, dossier attribute panel, compare attribute diff, /markets attribute market share chart. Command: `pnpm tsx scripts/enrich-attributes.ts --all` (script needs to be written). Cost: ~$14 one-time via Claude Haiku. 14K products × 10 tags avg × $0.0001 per tag.

2. **Run signal scores for all 14K products** — P0 · S · 30min
   Currently only 6,275 of 14,114 scored. Command: `pnpm signals`. Needed for: breakout alerts, trending page, velocity leaderboard, survival curve accuracy.

3. **Run relationship enrichment** — P1 · S · 30min
   Script exists: `pnpm enrich:relationships`. Populates product_alternatives via Claude Haiku competitor detection. ~$4 for existing 14K products. Unlocks: /explore graph edges, dossier alternatives, compare "attribute diff".

4. **Bulk news ingest** — P1 · M · 2h
   Only 73 press mentions. Run: `pnpm tsx --env-file=.env.local scripts/ingest-news.ts`. Hourly cron is accumulating going forward, but historical backfill needed. Unlocks: dossier press feed, funding extraction, home news strip.

5. **Create social_mentions table + Reddit/HN ingesters** — P1 · L · 1d
   Table not yet created. Needed for: dossier signal charts (platform breakdown), accurate velocity scores. Scripts designed, not built: scripts/ingest/reddit.ts, scripts/ingest/hackernews.ts.

6. **Fund funding_rounds via news extraction** — P2 · L · 1d
   Press mentions with event_type='funding' already being captured but not extracted to funding_rounds table. Build extraction step in news ingest pipeline: Claude detects funding → writes structured row to funding_rounds. Unlocks: dossier funding timeline, /markets leaderboard, /markets heatmap.

7. **Continue backfill to 25K+ products** — P2 · M · ongoing

Currently at 14,114. Target: 25K+ for meaningful lifecycle curves and category analytics. Single backfill process running. Watchdog script at scripts/watch-backfill.sh + run-backfill.sh will alert if it stalls.

8 **Seed functions table (~400 canonical values)** P2 M · 2h

The primary_function_id FK is in the schema but functions table is empty. Without it, /functions/[slug] pages can't exist. AI-bootstrap from PH categories + TAAFT taxonomy → human review → seed SQL. ~1hr human effort.

4-Layer Attribute Taxonomy

The "prism" concept — breaking every product into its component facets. This is what makes every unbuilt feature possible.

LAYER 1 — HIERARCHICAL (SINGLE-VALUE PER PRODUCT)

`category`	~15 top-level · "AI Tools", "Dev Tools"
`sub_category`	~100 mid-level · "Code Editors", "CRM"
`primary_function`	~400 leaf values · "ai-code-completion"

Leaf is FK to functions table. Category + sub-category auto-derive from parent chain.

LAYER 2 — MULTI-SELECT (PRODUCT_TAGS JOIN TAGS)

`capability`	collaborative, real-time, ai-assist, api-first...
`audience`	developers, enterprise, solopreneurs...
`deployment`	cloud, self-hosted, desktop-app, cli...
`pricing_model`	free, freemium, open-source, paid-only...
`tech_stack`	react, rust, python, go, typescript...
`ux_pattern`	keyboard-first, no-code, drag-drop, kanban...
`integration`	slack, notion, zapier, github, figma...
`compliance`	soc2, hipaa, gdpr, ccpa, iso27001...
`data_format`	markdown, json, csv, sql, images, video...
`business_model`	saas, marketplace, open-source-funded...

LAYER 3 — TASK DISCOVERY (TEXT[])

```
"write cold emails"
"summarize meeting notes"
"generate SQL queries"
"remove image backgrounds"
"transcribe video"
```

Powers /tools-to/[task] SEO pages. Massive long-tail traffic driver.

LAYER 4 — DIMENSIONAL SCORING (JSONB)

`ease_of_use`	4/5
`feature_depth`	5/5
`value_for_money`	3/5
`performance`	5/5
`integrations`	5/5

WHAT THIS UNLOCKS

FEATURE	REQUIRES	STATUS
/products faceted sidebar	L2 tags	BLOCKED
/functions/[slug] pages	L1 leaf + functions table	BLOCKED
/tools-to/[task] SEO	L3 task tags	BLOCKED
/attributes/[grp]/[val]	L2 tags	BLOCKED
Compare attribute diff	L2 tags	BLOCKED
Dossier attribute panel	L2 tags	PARTIAL
Attribute market share chart	L2 tags + launched_year	BLOCKED
Alternative finder (overlap)	L1 function + L2 tags	BLOCKED
Dimensional radar chart	L4 scores	BLOCKED
⌘K "tools that do X" search	L3 task tags	BLOCKED

SINGLE ACTION THAT UNLOCKS EVERYTHING

Writing and running **scripts/enrich-attributes.ts** — an attribute extraction pass over all 14K products using Claude Haiku — unblocks **all 10 features** in the table above. Estimated cost: ~$14 one-time. This is the highest-leverage task currently available.

AI-estimated at ingestion. Replace with G2/Capterra data Phase 4+.

Master Priority List — What to Build Next

Everything ordered by unlock leverage. P0 = blocks multiple features. Effort: S=<2h, M=half-day, L=full-day, XL=multi-day.

Markets completeness

#	TASK	UNLOCKS	PRI	EFFORT	STATUS
1	**Run pnpm signals (score all 14K products)** pnpm signals — already works, just needs to run	Breakout alerts, /trending, velocity charts	P0	S	TODO
2	**Build + run attribute extraction (enrich-attributes.ts)** Claude Haiku pass on 14K products: tags, task_search_tags, functionality_scores	All 10 taxonomy features, compare diff, attribute market share	P0	XL	NOT BUILT
3	**Build /trending page** is_breakout = TRUE products, sparklines, velocity chart	/trending route (missing from nav)	P0	S · 2h	NOT BUILT
4	**Run pnpm enrich:relationships** Script exists — Claude Haiku competitor detection. ~$4	Graph edges in /explore, dossier alternatives, compare	P1	S	SCRIPT READY
5	**Bulk news ingest** pnpm tsx scripts/ingest-news.ts — historical backfill	Press mentions feed, funding extraction, home news strip	P1	M	SCRIPT READY
6	**Build /products faceted sidebar** Accordion + Checkbox per tag_group, URL query params	Primary discovery UX	P1	L · 1d	NOT BUILT
7	**Add attribute market share chart to /markets** SQL query on product_tags × launched_year → recharts AreaChart stacked	The "components over time" feature	P1	M · 4h	NOT BUILT
8	**Dossier: add dimensional radar + signal 2×2** recharts RadarChart for functionality_scores, 2×2 signal charts	Dossier completeness (monetizable unit)	P1	M · 4h	NOT BUILT
9	**Seed functions table (~400 canonical values)** AI-bootstrap from PH + TAAFT → human review → seed SQL	/functions/[slug] pages	P1	M · 2h	NOT BUILT
10	**Build /functions/[slug] pages** Lifecycle chart, top-10 ranked, $-chart, full grid	Leaf taxonomy browsing, SEO	P1	L · 1d	NOT BUILT

11	**Build /tools-to/[task] pages** TAAFT-style, powered by task_search_tags[]	Long-tail SEO goldmine	P1	M · 3h	NOT BUILT
12	**Compare: add radar overlay + attribute diff + AI verdict** RadarChart overlay, 3-col diff, Claude Sonnet verdict	Compare page completion	P1	L · 1d	NOT BUILT
13	**Wire breakout alerts on home page** Simple query, component exists in wireframe	Home page completeness	P1	S · 1h	NOT BUILT
14	**Rename /products/[slug] → /dossier/[slug]** Update 6 files + add redirect	Branding + sub-brand "dossier" locked	P2	S · 2h	NOT DONE
15	**Build social_mentions table + Reddit/HN ingesters** New table, scripts/ingest/reddit.ts + hackernews.ts	Platform breakdown charts, accurate velocity	P2	XL · 2d	NOT BUILT
16	**Extract funding_rounds from press mentions** Add funding extraction step to news pipeline	Funding timeline, /markets heatmap + leaderboard	P2	L · 1d	NOT BUILT
17	**Insights: add Obsidian-style graph sidebar** Mini force-graph in right rail of /insights/[slug]	Insights page completeness	P2	L · 1d	NOT BUILT
18	**Build /attributes/[group]/[value] pages** Dynamic attribute slice pages, e.g. /attributes/deployment/self-hosted	SEO attribute pages	P2	M · 3h	NOT BUILT
19	**Build /companies/[slug]** Company profile: products, funding timeline, valuation chart	Company-level intelligence	P2	L · 1d	NOT BUILT
20	**Pre-death signals teaser + /graveyard death reason donut** Paid feature CTA. Death reason breakdown from product_graveyard table	Graveyard monetization hook	P2	M · 3h	NOT BUILT
21	**Add /markets category heatmap** Grid: category × year, color = VC $. Needs funding_rounds populated.	P2	L · 1d	NOT BUILT	
22	**Build /pricing page** Free / Pro / API tiers. Pre-death signals = Pro feature. API = paid.	Monetization starts	P3	M · 3h	NOT BUILT
23	**Wire user watchlist + alerts** user_saves + user_alerts tables exist (RLS). UI not wired.	Retention mechanic	P3	L · 1d	PLANNED

24	**Historical arc: seed pre-2010 products** Wikipedia + Computer History Museum + Archive.org. The real moat.	Historical differentiator. Evolution page depth.	P3	XL · weeks	PHASE 2